

UNIT-III – CLASSIFICATION TECHNIQUES IN DATA MINING

1. Definition and Applications of Classification

Definition

- Classification is a **supervised learning technique**
- It assigns data into **predefined classes or labels**

Example

- Student → Pass / Fail
 - Email → Spam / Not Spam
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Applications

- ☐ Spam detection
 - ☐ Medical diagnosis
 - ☐ Student performance prediction
 - ☐ Fraud detection
 - ☐ Customer segmentation
- ☐ **Key Point:** Uses labeled training data
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2. Decision Tree Classification

Definition

- A tree-based model used for decision making
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Structure

- Root node → top decision
- Internal node → attribute test

- Leaf node → class label
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ID3 Algorithm

- Uses **Information Gain**
- Selects attribute with highest gain

Steps:

1. Calculate entropy
 2. Compute information gain
 3. Choose best attribute
 4. Repeat recursively
-

C4.5 Algorithm

- Extension of ID3
 - Uses **Gain Ratio**
 - Handles:
 - Missing values
 - Continuous data
-

CART (Classification and Regression Tree)

- Uses **Gini Index**
 - Produces **binary tree**
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□ Comparison

Algorithm	Measure	Feature
ID3	Information Gain	Simple
C4.5	Gain Ratio	Improved
CART	Gini Index	Binary tree

3. Bayesian Classification (Naive Bayes)

Concept

- Based on **Bayes Theorem**

Formula

$$P(C|X) = (P(X|C) * P(C)) / P(X)$$

Assumption

- Features are **independent**
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Example

- Predict Pass based on:
 - Attendance
 - Marks
-

Advantages

- Simple and fast
 - Works well with large data
-

4. Rule-Based Classification

Concept

- Uses **IF-THEN rules**
-

Example

```
IF Attendance > 75 AND Marks > 50 → Pass  
ELSE → Fail
```

Characteristics

- Easy to understand
 - Human-readable
-

Rule Generation

- From decision trees
 - From algorithms like RIPPER
-

5. Model Evaluation Techniques

Confusion Matrix

Predicted + Predicted -	
Actual + TP	FN
Actual - FP	TN

Metrics

- Accuracy
 - Precision
 - Recall
 - F1-score
-

Cross-Validation

K-Fold Cross Validation

- Split data into K parts
 - Train on K-1
 - Test on remaining
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☐ **Advantage**

- Reduces overfitting
- Better evaluation

6. Improving Accuracy

A. Pruning

- Removes unnecessary branches in tree
- Reduces overfitting

B. Bagging (Bootstrap Aggregation)

- Multiple models trained on random samples
- Final result = average

C. Boosting

- Focuses on difficult samples
- Example: AdaBoost

☐ **Key Difference**

Technique	Idea
Pruning	Simplify model
Bagging	Reduce variance
Boosting	Reduce bias

7. Bayesian Belief Networks (Advanced)

Definition

- Graph-based probabilistic model
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Structure

- Nodes $\rightarrow$ variables
 - Edges $\rightarrow$ dependencies
-

Example

- Student performance depends on:
 - Attendance
 - Study hours
-

Advantages

- Handles uncertainty
 - Represents complex relationships
-

8. Neural Networks

Concept

- Inspired by human brain
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Structure

- Input layer
 - Hidden layers
 - Output layer
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Backpropagation

Steps

1. Forward pass (prediction)
 2. Calculate error
 3. Backward pass (update weights)
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☐ Use Cases

- Image recognition
 - Speech recognition
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9. Support Vector Machine (SVM)

Concept

- Finds best boundary (hyperplane)
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Key Idea

- Maximize **margin** between classes
-

Types

- Linear SVM
 - Non-linear (kernel trick)
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Advantages

- High accuracy
- Works well in high dimensions

10. Classification using Frequent Patterns

Concept

- Uses frequent itemsets for classification
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Example

- If:
 - {High Attendance, Good Marks} $\rightarrow$ Pass
-

Algorithms

- Associative classification
 - FP-Growth based classification
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Advantages

- Finds strong relationships
 - High interpretability
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☐ Final Summary

- Classification is widely used in **prediction problems**
 - Various models:
 - Decision Trees
 - Naive Bayes
 - Neural Networks
 - SVM
 - Evaluation and improvement techniques are essential
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□ Exam-Oriented Points

2-Mark

- Define classification
 - What is Naive Bayes?
 - What is SVM?
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16-Mark

- Explain Decision Tree algorithms
- Describe model evaluation techniques
- Discuss accuracy improvement methods